

Steven Lee Meisler

Curriculum Vitae

Steven is a developmental neuroinformaticist studying white matter maturation and its role in cognitive development and emerging psychopathology. As a NICHD K99/R00 postdoctoral fellow at the University of Pennsylvania, he leverages large-scale datasets and advanced analytic methods to characterize white matter developmental trajectories across infancy, childhood, and adolescence. His graduate work integrated multimodal neuroimaging (fMRI, DWI) with behavioral and cognitive assessments to investigate the neural bases of reading ability, disability, and intervention in children. Beyond his primary research, Steven is an active advocate for open and reproducible science. He moderates the NeuroStars.org support forum and teaches researchers how to process neuroimaging data using open-source, state-of-the-art tools. He received his bachelor's and master's degrees in bioengineering from the University of Pennsylvania (2017, 2018) and his PhD in Speech and Hearing Bioscience and Technology from Harvard University (2024).

Education

2019–2024 **PhD in Speech and Hearing Bioscience and Technology**, *Harvard University, Division of Medical Sciences*, Cambridge, MA
Secondary Field in Mind, Brain & Behavior
Dissertation: "White Matter Structural Correlates of Reading Abilities, Disabilities, and Intervention" - Advisor: John Gabrieli, PhD

2017–2018 **MSE in Bioengineering**, *University of Pennsylvania*, Philadelphia, PA, *3.79/4*
Thesis: "Evaluating the Effect of Intracranial EEG Data Cleaning on Univariate and Multivariate Classifier Performance" - Advisor: Michael Kahana, PhD

2013–2017 **BSE in Bioengineering**, *University of Pennsylvania*, Philadelphia, PA, *3.70/4 magna cum laude*
Minors in Mathematics, Jazz Studies, and Engineering Entrepreneurship

Research Positions

2024– **Postdoctoral Fellow**, *Lifespan Informatics & Neuroimaging Center*, University of Pennsylvania (PI: Theodore Satterthwaite, MD), NIH **K99/R00** and T32 Fellow
- Investigating white matter micro- and macrostructural trajectories in infants and toddlers, and how they are impacted by perinatal insult (e.g., low gestational age).
- Gauging the generalizability of white matter growth curves derived from research-quality datasets to data acquired clinically.
- Contributing to neuroimaging preprocessing software development (*QSIPrep*, *QSIRecon*, *XCP-D*).

Summer 2024 **Research Scientist Intern**, *Turing Medical*, St. Louis, MO
- Developed and tested neuroimaging software solutions for clinical practice.

2019–2024 **PhD Candidate**, *Gabrieli Lab*, MIT Brain and Cognitive Sciences (PI: John Gabrieli, PhD), NIH T32 Trainee and F31 Fellow
- Used advanced diffusion-weighted imaging models to study white matter microstructural correlates of reading in large pediatric cohorts ($>$ 1,000 participants).
- Studied longitudinal microstructural changes in white matter associated with response to reading intervention.
- Examined contributions of right hemispheric activity to reading improvement after intervention among dyslexic children.

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Funding

External

07/2026 – 06/2028 **Generalizable Growth Charts of Thalamocortical Development in Early Childhood**,
Funding Agency: NIH, NICHD (type: K99/R00; HD122318), Role: PI

9/2025 – 8/2026 **Hartwell Fellowship**,
Funding Agency: Hartwell Foundation

8/2024 – 8/2025 **Psychosis: A Convergent Neuroscience Perspective**,
Funding Agency: NIH, NIMH (type: T32; MH019112), Role: Trainee
(PI: Raquel E. Gur)

9/2023 – 05/2024 **Neurocognitive Mechanisms of Positive Intervention Response in Reading Disability**,
Funding Agency: NIH, NICHD (type: F31; HD111139), Role: PI

8/2019 – 8/2022 **Training in Speech and Hearing Sciences**,
Funding Agency: NIH, NIDCD (type: T32; DC000038), Role: Trainee
(PI: Gwenaelle S. Géléoc)

** Denotes Equal First Authorship; ** Denotes Equal Senior Authorship*

Published Articles (First Author)

1. **Meisler, S. L.,** Cieslak, M., Bagautdinova, J., Hendrickson, T. J., Pandhi, T., Chen, A. A., Hillman, N., Radhakrishnan, H., Salo, T., Feczko, E., Weldon, K. B., McCollum, r., Fayzullobekova, B., Moore, L. A., Sisk, L. M., Davatzikos, C., Huang, H., Avelar-Pereira, B., Caffarra, S., Chang, K., Cook, P. A., Flook, E. A., Gomez, T., Grotheer, M., Hagen, M. P., Huque, Z. M., Karipidis, I. I., Keller, A. S., Kruper, J., Luo, A. C., Macedo, B., Mehta, K., Mitchell, J. L., Pines, A. R., Pritschet, L., Rauland, A., Roy, E., Sevchik, B. L., Shafiei, G., Singleton, S. P., Stone, H. L., Sun, K. Y., Sydnor, V. J., Tong, T. T., Yablonski, M., Yeatman, J. D., Rokem, A., Shinohara, R. T., Fair, D. A., Satterthwaite, T. D., Highly replicable multisite patterns of adolescent white matter maturation. *Neuron*. ISSN: 0896-6273 (2026).
2. **Meisler*, S. L.,** Kubota*, E., Grotheer, M., Gabrieli, J. D. E., Grill-Spector, K., A practical guide for combining functional regions of interest and white matter bundles. *Frontiers in Neuroscience* **18,** 1385847 (2024).
3. **Meisler, S. L.,** Gabrieli, J. D. E., Christodoulou, J. A., White matter microstructural plasticity associated with educational intervention in reading disability. *Imaging Neuroscience* **2,** 1–18. ISSN: 2837-6056 (2024).
4. **Meisler, S. L.,** Gabrieli, J. D. E., Fiber-specific structural properties relate to reading skills in children and adolescents. *eLife* **11,** e82088. ISSN: 2050-084X (2022).
5. **Meisler, S. L.,** Gabrieli, J. D. E., A large-scale investigation of white matter microstructural associations with reading ability. *NeuroImage* **249,** 118909. ISSN: 1053-8119 (2022).
6. **Meisler, S. L.,** Kahana, M. J., Ezzyat, Y., Does data cleaning improve brain state classification? *Journal of Neuroscience Methods* **328,** 108421 (2019).

Articles Under Review / Revision (First Author)

1. **Meisler*, S. L.,** Salo*, T., Cieslak, M., Sevchik, B. L., Brook, J., Duhamel, G., Girard, O., Larsen, B., Pandey, S., Sisk, L. M., Soustelle, L., Sydnor, V. J., Johnson, P., Roalf, D. R., Shinohara, R. T., Taso**, M., Tisdall**, M. D., Satterthwaite**, T. D., An Open Test–Retest Dataset of Rapid Myelin Mapping Acquisitions. *bioRxiv* (2026).

Published Articles (Co-Author)

1. Cieslak, M., Irfanoglu, M. O., **Meisler, S. L.,** Salo, T., Raikes, A. C., Cook, P. A., Chung, A. W., Lee, E. G., Li, R., Li, X., Pecheva, D., Fair, D. A., Smyser, C. D., Harms, M. P., Landman, B. A., Wisnowski, J. L., Huang, H., Alexander, A. L., Satterthwaite,

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T. D., Diffusion MRI Processing in the HEALthy Brain and Child Development Study: Innovations and Applications. *Developmental Cognitive Neuroscience* (in production).

2. Sevchik, B. L., Shafiei, G., Murtha, K., Linguiti, S., Brodrick, L., Brook, J. B., Cieslak, M., Flook, E., Mehta, K., **Meisler, S. L.,** Ruparel, K., Rush, S., Salo, T., Singleton, S. P., Tong, T. T., Salunke, M., Bassett, D. S., Calkins, M. E., Elliott, M. A., Gur, R. E., Gur, R. C., Moore, T. M., Scott, J. C., Shinohara, R. T., Tisdall, M. D., Wolf, D. H., Roalf, D. R., Satterthwaite, T. D., An Open, Fully-processed, Longitudinal Data Resource to Study Brain Development and Transdiagnostic Executive Function. *Scientific Data* (2026).
3. Bagautdinova, J., Shafiei, G., Luo, A. C., Pecsok, M. K., Salo, T., Alexander-Bloch, A. F., Bassett, D. S., Gardner, M. E., Gur, R. E., Gur, R. C., Mackey, A. P., **Meisler, S. L.,** Misic, B., Moore, T. M., Roalf, D. R., Shinohara, R. T., Sydnor, V. J., Tong, T. T., Yeh, F.-C., Poldrack, R. A., Cieslak, M., Satterthwaite, T. D., Anatomical White Matter Tracts Span the Cortical Hierarchy to Support Cognitive Diversity. *Nature Human Behaviour,* 1–23 (2026).
4. Luo, A. C., **Meisler, S. L.,** Sydnor, V. J., Alexander-Bloch, A., Bagautdinova, J., Barch, D. M., Bassett, D. S., Davatzikos, C., Franco, A. R., Goldsmith, J., Gur, R. E., Gur, R. C., Hu, F., Jaskir, M., Kiar, G., Keller, A. S., Larsen, B., Mackey, A. P., Milham, M. P., Roalf, D. R., Shafiei, G., Shinohara, R. T., Somerville, L. H., Weinstein, S. M., Yeatman, J. D., Cieslak, M., Rokem, A., Satterthwaite, T. D., Two axes of white matter development. *Nature Communications* **17,** 1957 (2026).
5. Rauland, A., **Meisler, S. L.,** Alexander-Bloch, A. F., Bagautdinova, J., Baller, E. B., Gur, R. E., Gur, R. C., Luo, A. C., Moore, T. M., Popovych, O. V., Reetz, K., Roalf, D. R., Shinohara, R. T., Sotardi, S., Sydnor, V. J., Vossough, A., Eickhoff, S. B., Cieslak, M., Satterthwaite, T. D., White Matter Bundle Reconstruction From Single-Shell Diffusion Magnetic Resonance Imaging: Test–Retest Reliability and Predictive Capability Across Orientation Distribution Function Reconstruction Methods. *Human Brain Mapping* **46,** e70429 (2025).
6. Yang, Y.-F., Sólon Heinsfeld, A., Gondová, A., Vieira, B. H., Wang, Q., Mansour L. S., Li, X., Haugen, B., Bandettini, P., Bryant, A., Chakravary, M. M., Clarke, N., Clénet, B., Clucas, J. C., Dugré, M., Earl, E., Germani, E., Goodale, S. E., Gulban, O. F., Huber, R., Jeganathan, J., Joulot, M., Kai, J., Sitek, K. R., Koiso, K., Korbmacher, M., Maullin-Sapey, T., Maumet, C., McPherson, B. C., **Meisler, S. L.,** Mentch, J., Miedema, M., Moia, S., Nikolaidis, A., Taylor, P. A., Picard, M.-E., Pizzuti, A., Provins, C., Savary, E., Steinkamp, S. R., Taschler, B., Van Dyken, P. C., White, T., Yu, J.-C., Zou, Y., Haak, K. V., Proceedings of the OHBM Hackathon 2023. *Aperture Neuro* **5** (2025).
7. Sydnor, V. J., Bagautdinova, J., Larsen, B., Arcaro, M. J., Barch, D. M., Bassett, D. S., Alexander-Bloch, A. F., Cook, P. A., Covitz, S., Franco, A. R., Gur, R. E., Gur, R. C., Mackey, A. P., Mehta, K., **Meisler, S. L.,** Milham, M. P., Moore, T. M., Müller, E. J., Roalf, D. R., Salo, T., Schubiner, G., Seidlitz, J., Shinohara, R. T., Shine, J. M., Yeh, F.-C., Cieslak, M., Satterthwaite, T. D., A sensorimotor-association axis of thalamocortical connection development. *Nature Neuroscience,* 1–15 (2025).
8. Hur, K.-H., **Meisler, S. L.,** Yassin, W., Frederick, B. B., Kohut, S. J., Prefrontal-limbic circuitry is associated with reward sensitivity in nonhuman primates. *Biological Psychiatry* **96,** 473–485 (2024).
9. Wang, H.-T., **Meisler, S. L.,** Sharmarke, H., Clarke, N., Gensollen, N., Markiewicz, C. J., Paugam, F., Thirion, B., Bellec, P., Continuous evaluation of denoising strategies in resting-state fMRI connectivity using fMRIPrep and Nilearn. *PLOS Computational Biology* **20,** e1011942 (2024).
10. Poldrack*, R. A., Markiewicz*, C. J., **others**, The past, present, and future of the Brain Imaging Data Structure (BIDS). *Imaging Neuroscience* **2,** 1–19. ISSN: 2837-6056 (2024).
11. Decker, A. L., **Meisler, S. L.,** Hubbard, N. A., Bauer, C. C. C., Leonard, J., Grotzinger, H., Giebler, M. A., Torres, Y. C., Imhof, A., Romeo, R., Gabrieli, J. D. E., Striatal and Behavioral Responses to Reward Vary by Socioeconomic Status in Adolescents. *Journal of Neuroscience* **44** (2024).

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12. Marks, R. A., Pollack, C., **Meisler, S. L.,** D'Mello, A. M., Centanni, T. M., Romeo, R. R., Wade, K., Matejko, A. A., Ansari, D., Gabrieli, J. D. E., Christodoulou, J. A., Neurocognitive mechanisms of co-occurring math difficulties in dyslexia: Differences in executive function and visuospatial processing. *Developmental Science,* e13443 (2023).
13. Yu*, T., Cai*, L. Y., Torrisi, S., Vu, A. T., Morgan, V. L., Goodale, S. E., Ramadass, K., **Meisler, S. L.,** Lv, J., Warren, A. E., Englot, D. J., Cutting, L., Chang, C., Gore, J. C., Landman, B. A., Schilling, K. G., Distortion correction of functional MRI without reverse phase encoding scans or field maps. *Magnetic Resonance Imaging* **103,** 18–27 (2023).
14. Zhao, C., Tapera, T. M., Bagautdinova, J., Bourque, J., Covitz, S., Gur, R. E., Gur, R. C., Larsen, B., Mehta, K., **Meisler, S. L.,** Murtha, K., Muschelli, J., Roalf, D. R., Sydnor, V. J., Valcarcel, A. M., Shinohara, R. T., Cieslak, M., Satterthwaite, T. D., ModelArray: An R package for statistical analysis of fixel-wise data. *NeuroImage* **271,** 120037 (2023).
15. D'Mello, A. M., Frosch, I. R., **Meisler, S. L.,** Grotzinger, H., Perrachione, T. K., Gabrieli, J. D. E., Diminished Repetition Suppression Reveals Selective and Systems-Level Face Processing Differences in ASD. *Journal of Neuroscience* **43,** 1952–1962 (2023).
16. Richie-Halford, A., Cieslak, M., Ai, L., Caffarra, S., Covitz, S., Franco, A. R., Karipidis, I. I., Kruper, J., Milham, M., Avelar-Pereira, B., **others**, An analysis-ready and quality controlled resource for pediatric brain white-matter research. *Scientific Data* **9,** 1–27 (2022).
17. Edlow, B. L., Barra, M. E., Zhou, D. W., Foulkes, A. S., Snider, S. B., Threlkeld, Z. D., Chakravarty, S., Kirsch, J. E., Chan, S.-t., **Meisler, S. L.,** Bleck, T. P., Fins, J. J., Giacino, J. T., Hochberg, L. R., Solt, K., Brown, E. N., Bodien, Y. G., Personalized connectome mapping to guide targeted therapy and promote recovery of consciousness in the intensive care unit. *Neurocritical Care* **33,** 364–375 (2020).

Articles Under Review / Revision (Co-Author)

1. Raju, R. M., Davison, A. M., Rasheed, S., Boix, C. A., Abdelmoneum, M. M., Khera, N., Zhang, S., Liu, L., Fletcher, M. N. C., Barker, S. J. V., **Meisler, S. L.,** Milman, N. E. P., Pinals, R. L., Rodrigues-Amorim, D., Deik, A., Buenrostro, J. D., Bennett, D. A., Clish, C. B., Kellis, M., Tsai, L.-H., Choline rescues lipid and myelin deficits underlying male vulnerability to social isolation. *Under Review* (2026).
2. Macedo, B., Bagautdinova, J., **Meisler, S. L.,** Cieslak, M., Sisk, L. M., Davatzikos, C., Franco, A. R., Hettwer, M. D., Keller, A. S., Kiar, G., Luo, A. C., Mackey, A. P., Milham, M. P., Moore, T. M., Sydnor, V. J., Sun, K. Y., Yeh, F.-C., Barzilay, R., Fair, D. A., Shinohara, R. T., Alexander-Bloch, A., Satterthwaite, T. D., White matter reflects the childhood exposome. *bioRxiv* (2026).
3. McMaster, E. M., Rudravaram, G., Kim, M. E., Schwartz, T. M., Scholten, C., Yoon, J., Saunders, A. M., Hucke, A. T. S., Ramadass, K., Harriott, E. M., **Meisler, S. L.,** Vandekar, S. N., Newton, A., Smith, S. A., Sengupta, S., Humphreys, K. L., Osmundson, S., Moyer, D., Cutting, L. E., Landman, B. A., Harmonization mitigates diffusion MRI scanner effects in infancy: insights from the HEALthy Brain and Childhood Development (HBCD) study. *arXiv* (2026).
4. Takahesu Tabori, A. A., Azor, A., Marks, R. A., Kaminsky, A. J., Norton, R., Mastrangelo, C. M., Cardinaux, A. L., Wade, K., Thesken, H., **Meisler, S. L.,** Doyle, A. E., Braaten, E., Gabrieli, J. D. E., Christodoulou, J. A., Relations among home literacy activity, reading skills, and white-matter microstructure in school-aged children. *Under Review* (2025).

Invited Talks

1. **Meisler, S. L.** *The Reading Brain: Using Research to Understand Reading Acquisition and Disorders.* Howard University. Biology 444 Neuroscience Lecture. 2025.
2. **Meisler, S. L.** *BIDS Processing Workshop* Ohio State University. Led a Two-Day Workshop as part of the INCF-ReproNim Reproducible Neuroimaging Fellowship. 2024.
3. **Meisler, S. L.,** Gabrieli, J. D. E., *Fiber-Specific Properties Relate to Reading Ability.* Penn Lifespan Informatics and Neuroimaging Center Lab Meeting. 2022.

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4. D'Mello, A., Frosch, I. R., **Meisler, S. L.,** Grotzinger, H., Perrachione, T. K., Gabrieli, J. D., *Evidence for domain-specific neural adaptation reductions in autism spectrum disorder.* Society for Neuroscience. 2021.
5. **Meisler, S. L.,** Bodien, Y., Zhou, D., Edlow. B., *Comparing Brain Responses to Music and Language Stimuli to Classify Consciousness.* American Congress of Rehabilitation Medicine Conference. 2019.

Honors & Awards

2/2026-2/2028 **Career Development Institute in Psychiatry Fellowship**, University of Pittsburgh and Stanford University

9/2025-8/2026 **Hartwell Fellowship**, supporting early-stage biomedical research with potential to benefit children, Hartwell Foundation

9/2025-1/2026 **ABCD-ReproNim Teaching Fellow**, ReproNim: A Center for Reproducible Neuroimaging Computation

6/2024-06/2025 **Reproducible Neuroimaging Fellow**, ReproNim-International Neuroinformatics Coordinating Facility (INCF)

5/2024 **Ragnar & Margaret Naess Award** for exceptional musical talent and commitment to performance, MIT

8/2022 **Patrick J. McGovern Student Travel Award**, MIT

Summer 2022 **Selected for Neurohackademy 2022**

3/2022 **Certificate of Distinction in Teaching**, Harvard University

3/2020 **Mind Brain Behavior Graduate Student Award**, Harvard University

3/2019 **University Fellowship** *(declined)*, The Ohio State University

2017 ***magna cum laude***, University of Pennsylvania

2015 – 2017 **Dean's List**, University of Pennsylvania

Skills

Computing Python, Bash / Terminal, R, GitHub, High-Performance Computing, LaTeX, MATLAB

Methods DWI, fMRI, EEG Time Series Analysis, BIDS-apps, Longitudinal/Non-linear Modeling

Other Skills EPIC, REDCap, IRB, Jazz Vibraphone

Ad-Hoc Peer Reviewer

- ***Nature Human Behaviour, Nature Methods, Proceedings of the National Academy of Sciences, Journal of Neuroscience, Imaging Neuroscience, Biological Psychiatry, NeuroImage, Brain Structure & Function, npj Science of Learning, Neurobiology of Language, Brain Research Bulletin, Neuroinformatics, Neuropsychologia, Frontiers in Neuroscience***

Professional & Service Affiliations

2025–Present Trainee Committee Officer, Flux Congress

2025–Present Community Liaison, Open Science Special Interest Group, Organization for Human Brain Mapping

09/2024 Promoted to Moderator, NeuroStars support forum (`Neurostars.org`)

2019–Present Volunteer, Virtual Bedside Concerts, Mass General Brigham

2019–2021 Co-Chair, Arts & Neuroscience Cognition Task-Force, American Congress of Rehabilitation Medicine

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Teaching Experience

Teaching Fellow, ABCD-ReproNim Course

2025 **ABCD-ReproNim Course**

Supported a hybrid course on responsible and reproducible ABCD data analysis, including flipped-classroom instruction, data exercises, collaborative projects, and an in-person hackathon week focused on open-source neuroimaging workflows.

Invited Lectures

2025 **Howard University, BIOL 444: Neuroscience**

Gave a lecture on using multimodal neuroimaging to understand the cognitive processes underlying reading abilities and disabilities.

INCF-ReproNim Teaching Fellowship

2024 **Reproducible Neuroimaging Workshop**

Created and led a two-day workshop on reproducible, open source, end-to-end neuroimaging workflows. Taught best practices in data curation, pre- and post-processing for functional and diffusion MRI.

Teaching Fellow, Harvard University

Falls 2021-2023 **How Music Works: Engineering the Acoustical World**

Led sections and supported hands-on projects for an interdisciplinary engineering course using music and acoustics to introduce concepts in physics, mathematics, signal processing, digital audio, and design.

Teaching Assistant, University of Pennsylvania

Fall 2017 **Differential Equations & Linear Algebra**

Supported instruction in matrices, linear systems, vector spaces, linear transformations, and systems of ordinary differential equations, with emphasis on engineering applications.

Fall 2016 & 2017 **Bioengineering Signals & Systems**

Supported instruction in continuous and discrete signals, linear time-invariant systems, Fourier analysis, filtering, sampling, and biomedical applications including ECG, blood pressure, auditory coding, cochlear implants, and imaging.

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